ACADEMIC CENTRAL

PRACTICE EXAMINATION

NC(V) LEVEL 2

ELECTRONICS

PAPER 1 · PRACTICE A

MARKS: 100 **TIME: 3 hours**

NC(V) Information Technology and Computer Science, Level 2. Section A: objective questions (30) · Section B: safety, electrical principles, Ohm's law and power, series and parallel circuits, components and measuring instruments (70).

Written by Academic Central for practice, following the NC(V) subject guidelines for NC(V) Level 2. It is not a Department of Higher Education and Training examination paper.

INSTRUCTIONS AND INFORMATION

1. Answer ALL the questions.
2. Read ALL the questions carefully.
3. Number the answers according to the numbering system used in this question paper.
4. Show ALL formulae, substitutions and calculations. Round off answers to TWO decimal places.
5. Write units with all answers.
6. A non-programmable calculator may be used.
7. Write neatly and legibly.

SECTION A

QUESTION 1

1.1 Various options are given as possible answers to the following questions. Choose the answer and write only the letter (A–D) next to the question number (1.1.1–1.1.10).

1.1.1 The unit of electrical resistance is the …

A volt. B ampere. C watt. D ohm.

1.1.2 Which ONE of the following materials is a good insulator?

A Copper B Rubber C Aluminium D Salt water

1.1.3 An ammeter is connected … with the load.

A in parallel B across C diagonally D in series

1.1.4 The particle with a negative charge in an atom is the …

A proton. B electron. C neutron. D nucleus.

1.1.5 In a parallel circuit the … is the same across all branches.

A current B resistance C voltage D power

1.1.6 A resistor with the colour bands brown, black, red has a value of …

A 1 kΩ. B 10 Ω. C 100 Ω. D 10 kΩ.

1.1.7 A component that allows current to flow in one direction only is a …

A capacitor. B diode. C resistor. D fuse.

1.1.8 The unit of capacitance is the …

A henry. B coulomb. C hertz. D farad.

1.1.9 The frequency of the mains supply in South Africa is …

A 50 Hz. B 60 Hz. C 230 Hz. D 100 Hz.

1.1.10 0,005 A is the same as …

A 0,5 mA. B 50 mA. C 5 mA. D 500 mA.

(10)

1.2 Indicate whether the following statements are TRUE or FALSE. Choose the answer and write only 'true' or 'false' next to the question number (1.2.1–1.2.10).

1.2.1 Current is the flow of electrons through a conductor.

1.2.2 Adding more resistors in series decreases the total resistance.

1.2.3 A fuse protects a circuit by melting when the current is too high.

1.2.4 A battery supplies alternating current (AC).

1.2.5 Power is calculated as voltage multiplied by current.

1.2.6 A voltmeter has a very low internal resistance.

1.2.7 The anode of a light-emitting diode (LED) is normally the longer leg.

1.2.8 Static electricity can damage electronic components such as memory chips.

1.2.9 Semiconductors such as silicon are used to make transistors.

1.2.10 In a series circuit, if one bulb blows, the other bulbs stay on.

(10)

1.3 Choose a term from COLUMN B that matches a description in COLUMN A. Write only the letter (A–J) next to the question number (1.3.1–1.3.10).

	COLUMN A		**COLUMN B**
1.3.1	Stores electrical charge	A	multimeter
1.3.2	Opposes the flow of current	B	transistor
1.3.3	Electrical pressure that pushes current around a circuit	C	anti-static strap
1.3.4	Rate at which electrical energy is used	D	capacitor
1.3.5	Instrument that measures voltage, current and resistance	E	power
1.3.6	Resistor whose value changes with light	F	voltage
1.3.7	Electronically switches or amplifies a signal	G	earth leakage unit
1.3.8	A path with zero resistance that bypasses the load	H	LDR
1.3.9	Wrist strap used to prevent static discharge	I	short circuit
1.3.10	Device that switches off the supply when current leaks to earth	J	resistor

(10)

TOTAL SECTION A: [30]

SECTION B

QUESTION 2: SAFETY AND ELECTRICAL PRINCIPLES

2.1 Give FOUR safety rules that must be followed when working on electronic equipment in a workshop. (4)

2.2 Explain the difference between a conductor and an insulator. Give ONE example of each. (4)

2.3 Explain the difference between direct current (DC) and alternating current (AC). Draw a simple waveform of each. (6)

2.4 Name TWO effects of an electric current and give ONE practical example of each. (4)

2.5 What should you do first if you find a person who is receiving an electric shock? (2)

[20]

QUESTION 3: OHM'S LAW AND POWER

3.1 State Ohm's law in words. (2)

3.2 A cellphone charger supplies 5 V to a circuit with a resistance of 2,5 Ω. Calculate the current. (3)

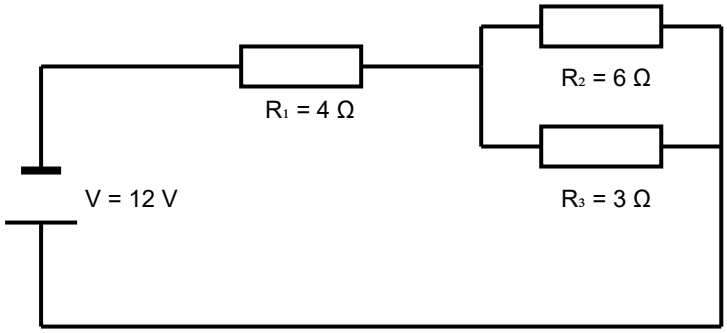
3.3 A soldering iron draws 0,25 A from a 230 V supply. Calculate: (a) its resistance; (b) its power. (6)

3.4 A 60 W lamp is switched on for 5 hours a day for 30 days. Calculate the energy used in kWh and the cost if electricity costs R3,20 per kWh. (4)

[15]

QUESTION 4: SERIES AND PARALLEL CIRCUITS

4.1 Study the circuit below and answer the questions.



R_2 and R_3 are connected in parallel, and this combination is in series with R_1.

4.1.1 Calculate the equivalent resistance of R_2 and R_3. (3)

4.1.2 Calculate the total resistance of the circuit. (2)

4.1.3 Calculate the total current drawn from the supply. (2)

4.1.4 Calculate the voltage across R_1. (2)

4.1.5 Calculate the voltage across R_3. (2)

4.1.6 Calculate the current through R_2. (2)

4.1.7 Calculate the total power delivered by the supply. (2)

4.2 Give TWO advantages of connecting the lights in a house in parallel rather than in series. (2)

4.3 Three 1,5 V cells are connected (a) in series and (b) in parallel. What is the total voltage in each case? (2)

[19]

QUESTION 5: COMPONENTS AND MEASURING INSTRUMENTS

5.1 Give the value and tolerance of a resistor with the colour bands yellow, violet, orange, gold. (3)

5.2 Draw the circuit symbols for: (a) a diode; (b) a capacitor (non-polarised); (c) a fuse; (d) an LED. (4)

5.3 Explain why a resistor is connected in series with an LED. (2)

5.4 Explain how you would use a multimeter to test whether a fuse is blown. (3)

5.5 Give TWO uses of capacitors in electronic circuits. (2)

5.6 Name ONE component inside a computer that uses (a) a transistor and (b) a capacitor. (2)

[16]

TOTAL SECTION B: [70]

TOTAL: 100

END OF QUESTION PAPER